Docking Universal - Docking Study Report

Target: 1HVR | Ligands: 2 compounds

1. Bound-ligand control: XK2

This retrospective control tests whether the protocol reproducibly recovers the experimental pose of XK2. PASS requires sampling, ranking, and independent-seed criteria to pass. It supports use of the selected protocol for this target, but does not establish affinity or prospective pose accuracy.

Control metric	Result
Overall protocol status	PASS
Sampling control	PASS
Ranking control	PASS
Independent-seed requirement	PASS - 5 observed; 5 required
RMSD threshold	2.0 A
Best sampled RMSD	0.5788 A
Top-ranked pose RMSD	0.6578 A
Top-ranked docking score	-11.801 kcal/mol

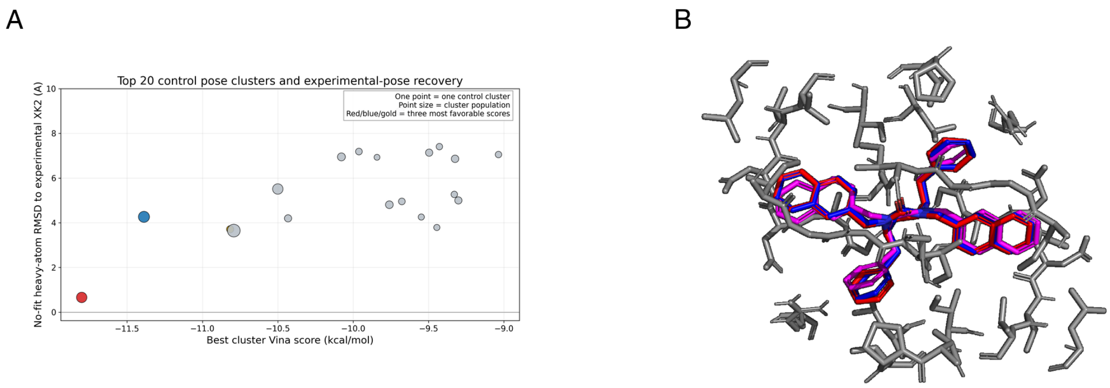

Figure 1. Retrospective control performance and pose recovery. (A) Control-cluster Vina score versus symmetry-aware, no-fit heavy-atom RMSD to experimental XK2 in the receptor coordinate frame. (B) Experimental ligand (magenta), lowest-energy pose (red), and lowest-RMSD pose (blue) superimposed in that frame; receptor residues within 5 A of the displayed ligands are gray.

Control interaction diagrams

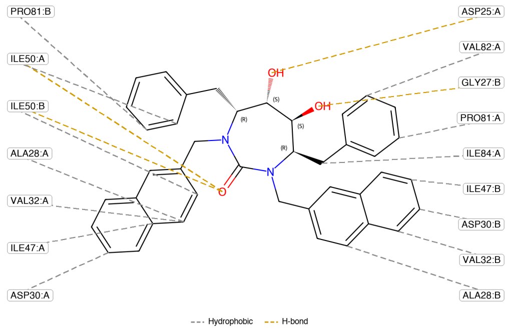

Figure 2. SDF-aware PLIP interaction diagram. Experimental XK2 pose used as the control reference. Ligand chemistry comes from the retained SDF and interaction calls come from the retained PLIP XML.

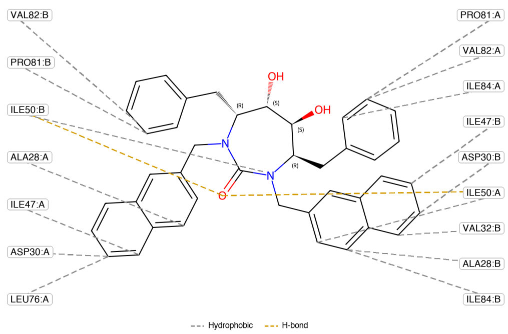

Figure 3. SDF-aware PLIP interaction diagram. Globally lowest-energy redocked pose (Vina score -11.801 kcal/mol; RMSD 0.6578 A). Ligand chemistry comes from the retained SDF and interaction calls come from the retained PLIP XML.

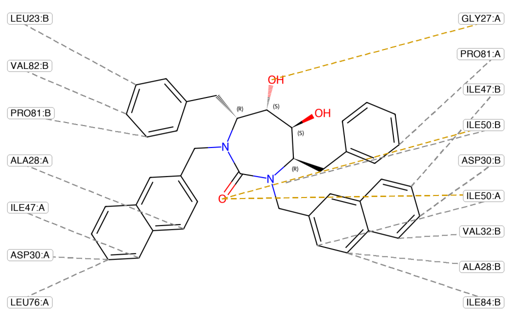

Figure 4. SDF-aware PLIP interaction diagram. Globally lowest-RMSD redocked pose (RMSD 0.5788 A). Ligand chemistry comes from the retained SDF and interaction calls come from the retained PLIP XML.

Approved protocol

Parameter	Selected value
Engine	vina
Tier	repeatability
Exhaustiveness	16
Modes per job	15
Conformers per state	3
Independent seeds	5
Charge model	gasteiger
pH	7.4
Runtime	18.9 min
Calibration jobs	15

2. Docking results

Ligand: Nevirapine

Docking used the approved target-matched protocol shown above. More favorable Vina scores are more negative; RMSD and cluster population are separate measures.

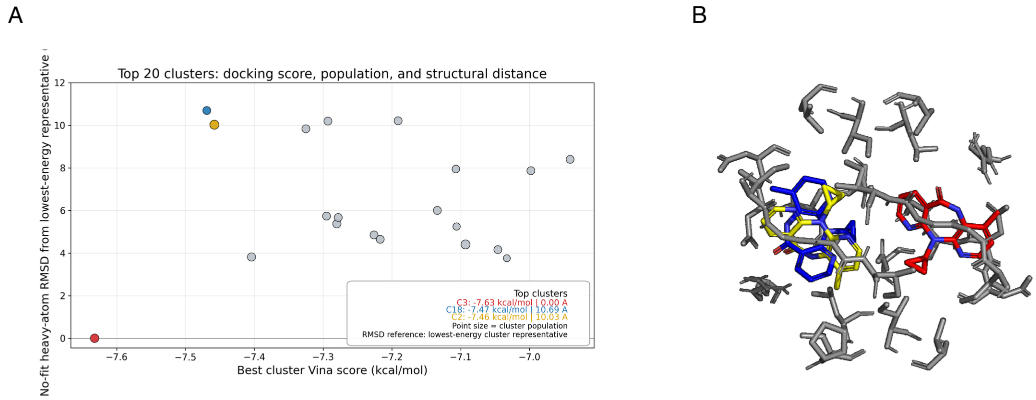

Figure 5. Docking pose-cluster analysis for Nevirapine. (A) Docking score versus symmetry-aware, no-fit heavy-atom RMSD from the lowest-energy cluster representative in the receptor coordinate frame; point size denotes cluster population. (B) Representative structures from the highlighted clusters, using matching cluster colors.

Top-ranked 3D cluster snapshots

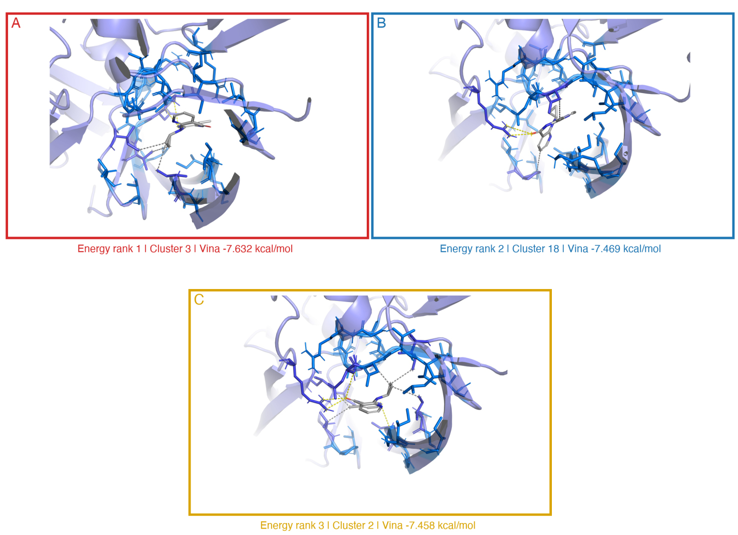

Figure 6. Three-dimensional interaction snapshots for Nevirapine. Shown are 3 energy-ranked distinct cluster representatives, ordered by Vina score. Red, blue, and gold match the highlighted clusters in Figure 5. These views support structural inspection; docking score rank does not establish pose correctness.

Top-ranked pose interaction diagrams

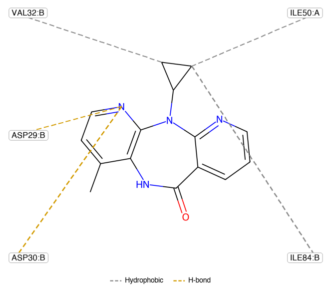

Figure 7. SDF-aware PLIP interaction diagram for Nevirapine, energy rank 1, cluster 3 (Vina score -7.632 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

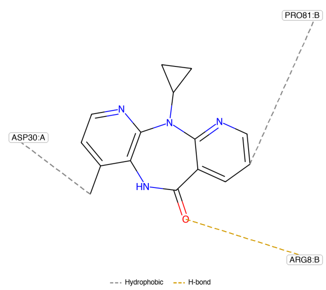

Figure 8. SDF-aware PLIP interaction diagram for Nevirapine, energy rank 2, cluster 18 (Vina score -7.469 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

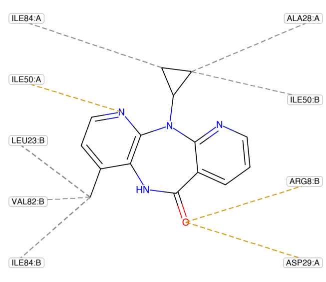

Figure 9. SDF-aware PLIP interaction diagram for Nevirapine, energy rank 3, cluster 2 (Vina score -7.458 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

Ranked docking clusters

Rank	Cluster	Best score	Median score	Poses	Seeds
1	**Cluster 3**	-7.632	-7.618	7	5
2	**Cluster 18**	-7.469	-7.458	5	5
3	**Cluster 2**	-7.458	-7.439	9	5
4	Cluster 5	-7.404	-7.361	6	5
5	Cluster 17	-7.325	-7.323	5	5
6	Cluster 12	-7.295	-7.284	5	5
7	Cluster 11	-7.293	-7.253	5	5
8	Cluster 4	-7.28	-7.267	6	5
9	Cluster 16	-7.278	-7.267	5	5
10	Cluster 15	-7.226	-7.225	5	5
11	Cluster 10	-7.217	-7.208	5	5
12	Cluster 9	-7.191	-7.177	5	5

Rank	Cluster	Best score	Median score	Poses	Seeds
13	Cluster 8	-7.134	-7.121	5	5
14	Cluster 23	-7.107	-7.096	4	4
15	Cluster 26	-7.106	-7.095	3	3
16	Cluster 1	-7.093	-7.021	9	5
17	Cluster 7	-7.046	-7.042	5	5
18	Cluster 33	-7.033	-7.008	2	2
19	Cluster 6	-6.998	-6.984	5	5
20	Cluster 14	-6.941	-6.932	5	5

Interpretation and limitations

Docking scores are ranking estimates, not measured binding free energies. Rigid-receptor docking does not model induced fit. Cluster population and seed support describe computational convergence, not biological correctness. Protonation, tautomer, receptor preparation, and box choices can affect results. Experimental validation remains necessary.

Ligand: Rilpivirine

Docking used the approved target-matched protocol shown above. More favorable Vina scores are more negative; RMSD and cluster population are separate measures.

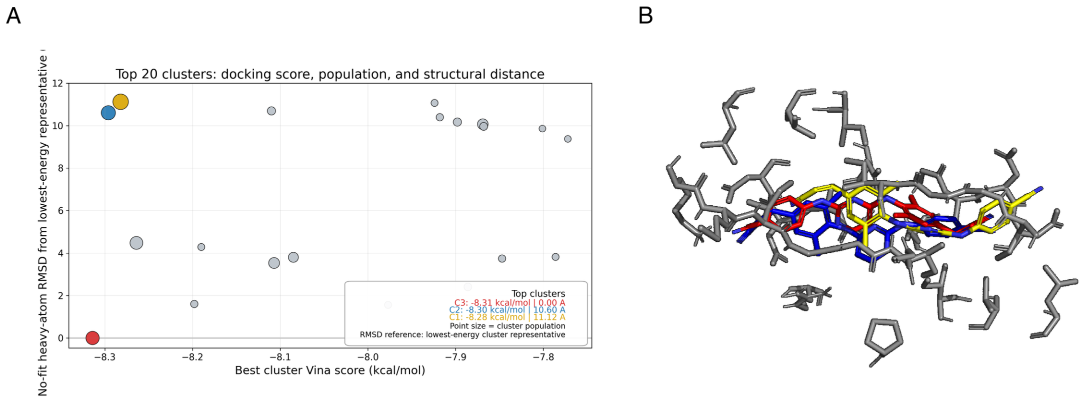

Figure 10. Docking pose-cluster analysis for Rilpivirine. (A) Docking score versus symmetry-aware, no-fit heavy-atom RMSD from the lowest-energy cluster representative in the receptor coordinate frame; point size denotes cluster population. (B) Representative structures from the highlighted clusters, using matching cluster colors.

Top-ranked 3D cluster snapshots

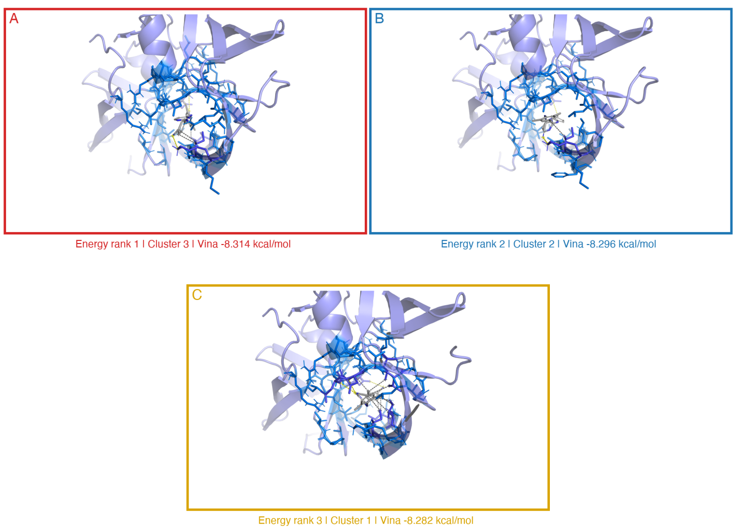

Figure 11. Three-dimensional interaction snapshots for Rilpivirine. Shown are 3 energy-ranked distinct cluster representatives, ordered by Vina score. Red, blue, and gold match the highlighted clusters in Figure 10. These views support structural inspection; docking score rank does not establish pose correctness.

Top-ranked pose interaction diagrams

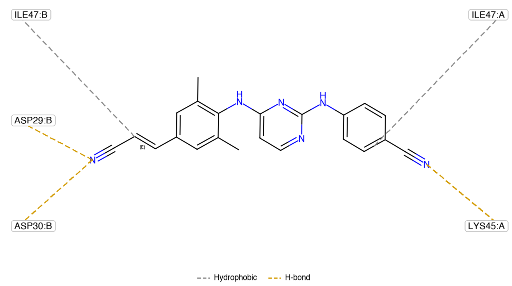

Figure 12. SDF-aware PLIP interaction diagram for Rilpivirine, energy rank 1, cluster 3 (Vina score -8.314 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

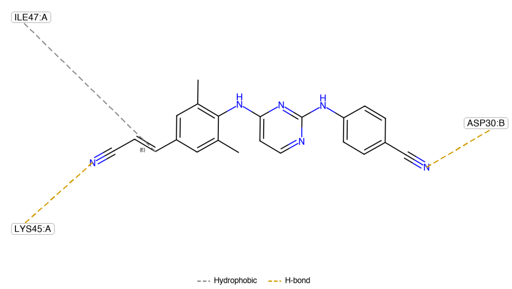

Figure 13. SDF-aware PLIP interaction diagram for Rilpivirine, energy rank 2, cluster 2 (Vina score -8.296 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

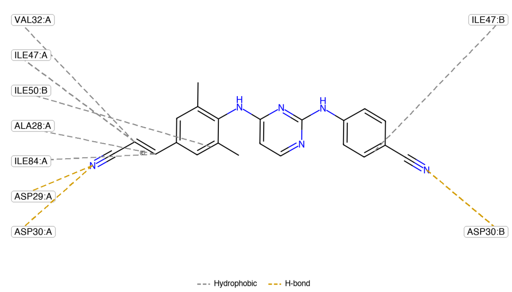

Figure 14. SDF-aware PLIP interaction diagram for Rilpivirine, energy rank 3, cluster 1 (Vina score -8.282 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

Ranked docking clusters

Rank	Cluster	Best score	Median score	Poses	Seeds
1	**Cluster 3**	-8.314	-8.232	30	5
2	**Cluster 2**	-8.296	-8.181	35	5
3	**Cluster 1**	-8.282	-8.188	44	5
4	Cluster 4	-8.264	-8.016	27	5
5	Cluster 20	-8.198	-8.196	3	3
6	Cluster 15	-8.19	-8.153	2	2

Rank	Cluster	Best score	Median score	Poses	Seeds
7	Cluster 18	-8.11	-7.743	6	4
8	Cluster 13	-8.107	-7.961	17	5
9	Cluster 14	-8.085	-7.895	13	5
10	Cluster 16	-7.977	-7.881	2	1
11	Cluster 5	-7.924	-7.874	2	2
12	Cluster 8	-7.918	-7.842	3	3
13	Cluster 17	-7.898	-7.832	6	3
14	Cluster 7	-7.886	-7.808	4	4
15	Cluster 6	-7.869	-7.76	16	4
16	Cluster 9	-7.868	-7.777	5	4
17	Cluster 19	-7.847	-7.692	3	1
18	Cluster 23	-7.801	-7.801	1	1
19	Cluster 11	-7.786	-7.669	2	2
20	Cluster 10	-7.772	-7.772	1	1

Interpretation and limitations

Docking scores are ranking estimates, not measured binding free energies. Rigid-receptor docking does not model induced fit. Cluster population and seed support describe computational convergence, not biological correctness. Protonation, tautomer, receptor preparation, and box choices can affect results. Experimental validation remains necessary.

3. Reproducibility, software, and references

Docking scores and poses were produced by AutoDock Vina. PLIP supplied rule-based protein-ligand interaction calls; the retained PLIP XML is the authoritative interaction record. RDKit supplied the retained molecular graph handling and symmetry-aware, no-fit heavy-atom RMSD matrix used by Butina clustering. The clustering cutoff was 2.0 A. If only one cluster is present, its lowest-energy member is reported as the sole representative. PyMOL produced the 3D molecular panels; ReportLab assembled this PDF.

Software versions used for this report

Result element	Software	Version used
Workflow	Docking Universal	0.4.0
Docking scores and poses	AutoDock Vina	AutoDock Vina v1.2.7 (conda environment docking-universal-vina)
Docking parameterization	Meeko	0.6.1
Protonation/conformer preparation	MolScrub	0.2.2
Molecular graph, RMSD, clustering	RDKit	2023.9.6
Molecular conversion/PLIP backend	Open Babel	3.1.0
Interaction calls	PLIP	2.3.1
3D rendering	PyMOL	3.0.0
Plots	Matplotlib	3.9.4
PDF generation	ReportLab	5.0.0
Runtime	Python	3.9.23

Scientific and software references

1. Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. J Chem Inf Model. 2021;61:3891-3898. https://doi.org/10.1021/acs.jcim.1c00203

2. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. https://doi.org/10.1021/acs.jcim.5c02271

3. Salentin S, Schreiber S, Haupt VJ, Adasme MF, Schroeder M. PLIP: fully automated protein-ligand interaction profiler. Nucleic Acids Res. 2015;43:W443-W447. https://doi.org/10.1093/nar/gkv315

4. Butina D. Unsupervised Data Base Clustering Based on Daylight's Fingerprint and Tanimoto Similarity: A Fast and Automated Way To Cluster Small and Large Data Sets. J Chem Inf Comput Sci. 1999;39:747-750. https://doi.org/10.1021/ci9803381

5. O'Boyle NM, Banck M, James CA, Morley C, Vandermeersch T, Hutchison GR. Open Babel: An open chemical toolbox. J Cheminform. 2011;3:33. https://doi.org/10.1186/1758-2946-3-33

6. RDKit: Open-source cheminformatics software. https://www.rdkit.org/

7. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. https://www.pymol.org/support.html